

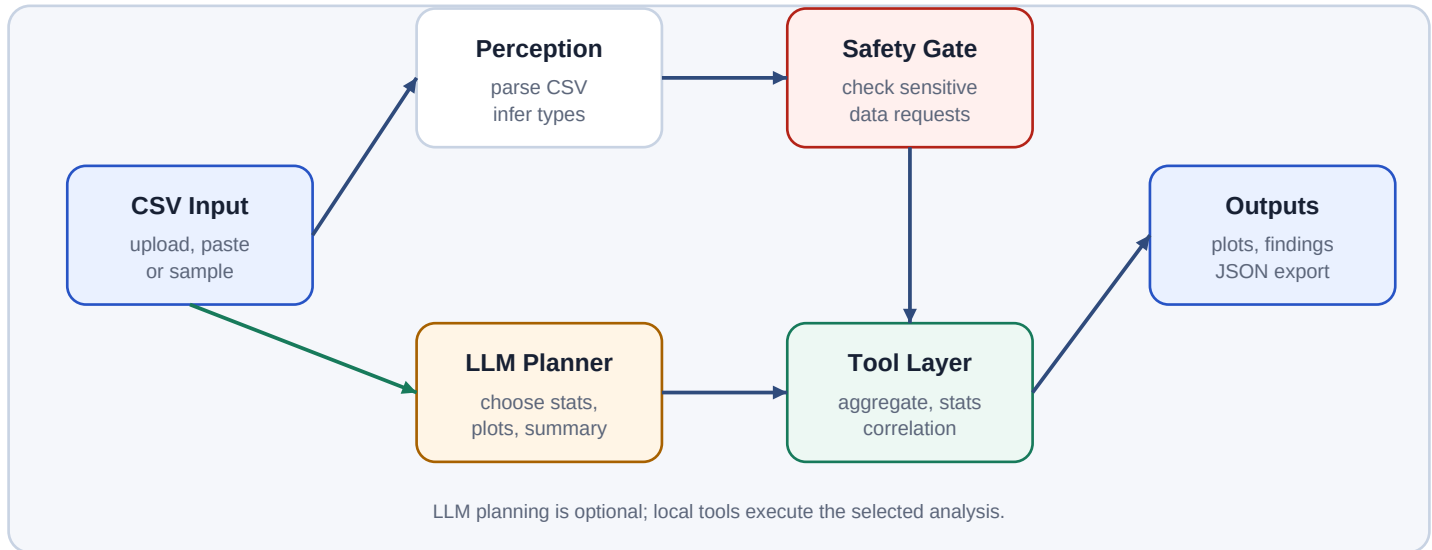
DataLens Agent Report

CSV analysis agent that chooses steps, creates plots, and summarizes findings

GitHub repository: <https://github.com/your-username/datalens-agent> (replace with the uploaded repository URL before submission)

Prototype: A browser-based data analysis agent that accepts pasted or uploaded CSV data and runs an end-to-end analysis loop.

System Design



Design Explanation

DataLens follows a perceive-decide-act agent loop. The user provides a CSV and an analysis goal. The perception module parses the CSV, infers numeric, categorical, and date columns, and checks missing values. The safety gate blocks requests that appear to involve secrets or sensitive identifiers. The optional LLM planner selects suitable analysis steps from the dataset shape and user goal; if it is unavailable, deterministic rules select the same local tools. The tool layer executes schema inspection, numeric profiling, category aggregation, relationship plotting, and summary generation.

Agent Concepts Used

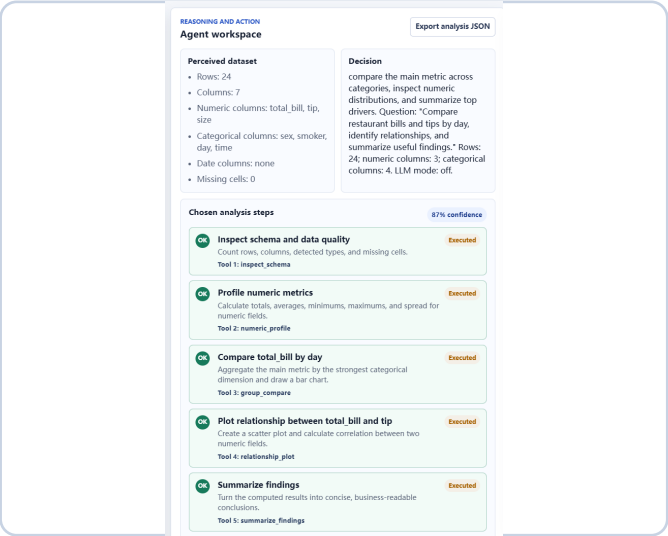
- **Perception:** parses CSV rows and infers column types.
- **Decision making:** optionally uses an LLM planner, with deterministic fallback, to choose analysis steps.
- **Action:** executes local analysis tools, draws charts on canvas, and can use the LLM to rewrite the final summary.
- **Memory:** records loaded datasets, LLM/fallback status, and executed tool runs in localStorage.
- **Safety:** refuses analysis requests involving secrets or sensitive identifiers.

Assignment 2

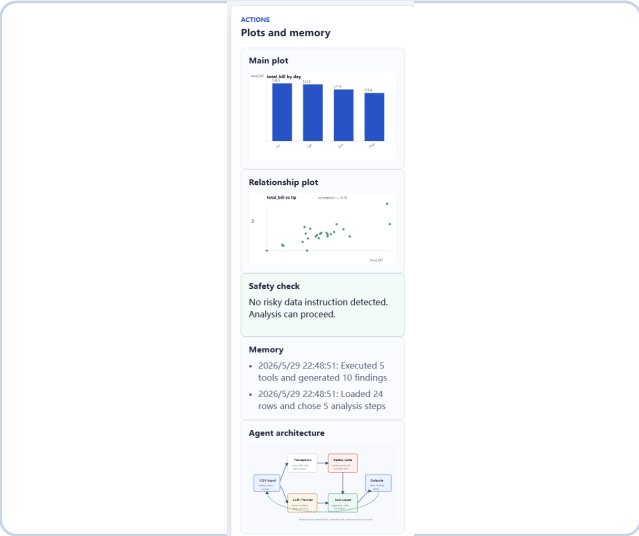
Screenshots and System Behavior

Evidence of CSV perception, step choice, plotting, execution, and summary

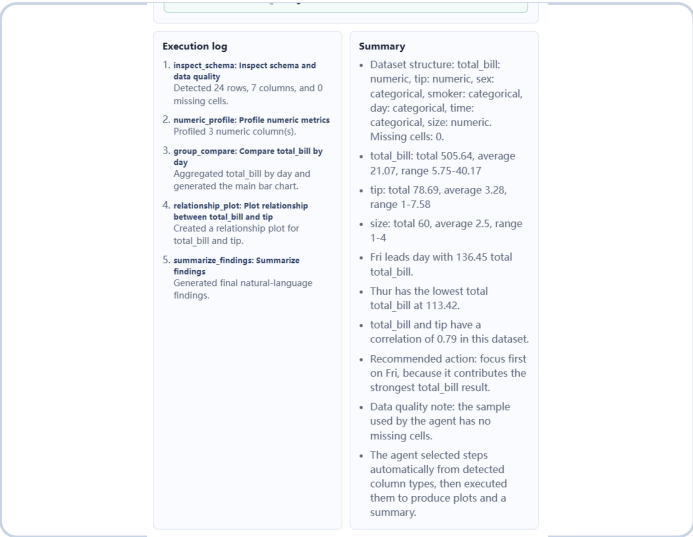
The screenshots below show the DataLens agent running a small public tips CSV sample. The agent chooses its analysis plan automatically from the detected fields.



1. Perception and decisions: the agent detects 24 rows with numeric and categorical fields, then chooses analysis steps.



2. Plot actions: the agent creates a category comparison bar chart and a relationship scatter plot with correlation.



3. Summary: findings explain top categories, numeric profiles, correlation, and recommended business action.

How It Works

- **Input:** user uploads, pastes, or loads a sample CSV.
- **Perception:** CSV parser builds records and infers data types.
- **Decision:** the agent chooses profiling, aggregation, plotting, and summary steps.
- **Execution:** each selected tool runs and writes to the execution log.
- **Output:** charts, findings, memory, and JSON export make the result reproducible.